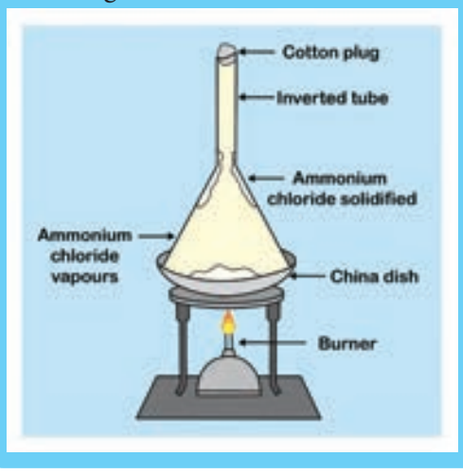
evaporation, boiling, condensation, melting and freezing are all physical processes

ACTIVITY 5

Sublimation

Take some camphor in a porcelain dish and cover it with a clean glass funnel. Close the mouth of the funnel with small amount of cotton wool. Heat the contents in the dish. can you see that camphor changes into vapour state without becoming liquid.

Ammonium chloride is another substance that undergoes sublimation.



3.3.6 Crystallization

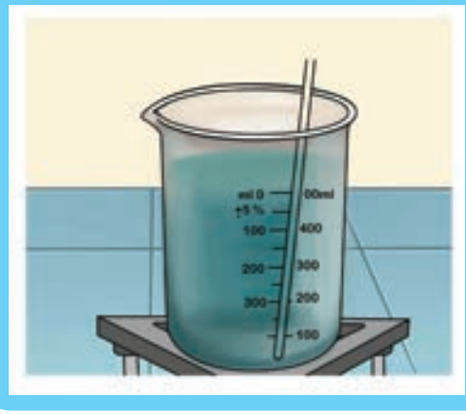
Though not mentioned earlier, crystallization is also a special form of physical change. The soluble impurities get removed from certain solids by crystallization. The process of cooling a hot, concentrated solution of a substance to obtain crystals is called crystallization.

We also know that sea-water contains salts dissolved in it and the salt can be separated from sea-water by the process of evaporation. The process of evaporation is not a good technique because the soluble impurities do not get removed in the process of evaporation.

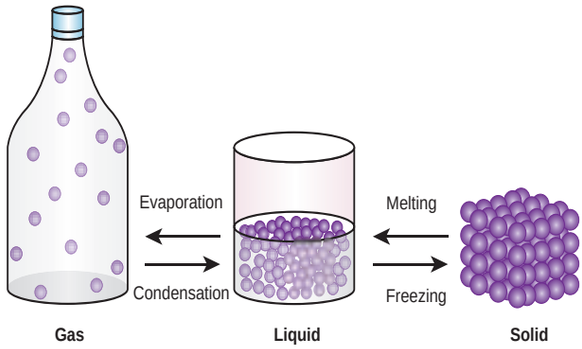
ACTIVITY 6

Crystallizing copper sulphate

Take about 100ml of water in a beaker. Heat the water over a burner till it boils. Add impure copper sulphate to the hot water with constant stirring. Continue to add copper sulphate till the solution takes up the added copper sulphate, that is, the added copper sulphate will not dissolve anymore. Filter the contents on a glass plate and allow it to cool. You could see crystals of copper sulphate in a few hours.



Further the crystals of salts obtained by the process of evaporation are small. The shape of crystals cannot be seen clearly. So the solid substances are usually purified by the process of crystallization. Large crystals of pure substances can be obtained from their solutions by the process of crystallization. **Crystallization is a method of separation as well as a method of purification.**



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